USRP X310 Data Recording Report

Rubem Vasconcelos Pacelli[1]

[1]Aeronautics Institute of Technology (ITA)

Last update: October 22, 2026

Abstract

This report describes the data recording process using the USRP X310 device. This reciver is placed in a fixed position in São José dos Campos, SP, Brazil, in the GPS/GNSS Laboratory at Aeronautics Institute of Technology (ITA)[1]. The present report is updated as new recordings are made and new information is available.



Figure 1: Location of the GNSS antenna.

[1]https://github.com/ita-gnss-lab

Contents

I Recording overview **3**

1 Methodology **3**

2 Recording calendar **4**

3 Validating the recordings with GNSS-SDR **4**
3.1 Signal source 5
3.2 Signal conditioner 6
3.3 Acquisition 6
3.4 Tracking 6
3.5 Telemetry decoder 6
3.6 Observables 7
3.7 PVT 7

II Recording reports **8**

4 Records on 08/10/2026 **8**
4.1 First record 8

5 Records on 13/10/2026 **9**
5.1 First record 9
5.2 Second record 10

6 Records on 22/10/2026 **11**
6.1 First record 11
6.1.1 Signal source 11
6.1.2 Signal conditioner 12
6.1.3 Acquisition 13
6.1.4 Tracking 14
6.1.5 Telemetry decoder 15
6.1.6 Observables 16
6.1.7 PVT 17

Part I
Recording overview

This part introduces an overview os the recordings presented in this report. It includes the methodology used to record the data, the calendar of the recorded days, the configuration of the USRP X310, and the processing of the recordings with GNSS-SDR.

1 Methodology

The recording is computed automatically by a Linux service which is driven by the following guidelines:

- Everyday, the program will start within the time rage [4PM, 6PM]
- It will sense whether the USRP is detectable via `uhd_find_devices --args=''addr=192.168.10.2''`.
- If not found, it will exit silently and will sense again in intervals of 15 min.
- Once found, two recordings are made, each with duration of 5 minutes (300 s). The first recording happens right after the USRP is detected, and the second recording is made 30 min after the first one. After that, no more recordings will be made for that day.
- The two recordings of a day use different signal pairs:
  - First recording: channel 0 (RX0) GPS L1 C/A (1575.42 MHz), 4 MS/s; channel 1 (RX1) GPS L5 (1176.45 MHz), 20 MS/s; both 25 dB of gain, signed 8-bit I/Q (2 bytes per complex sample);
  - Second recording: channel 0 (RX0) GPS L1 C/A (1575.42 MHz); channel 1 (RX1) GPS L2C (1227.60 MHz); both 4 MS/s, 25 dB of gain, signed 16-bit I/Q (4 bytes per complex sample).
- For each recording a SigMF archive file is created, with the interleaved I/Q samples + metadata, like number of channels (dual channel recording), frequencies, sampling rate, gain, sample type, etc. Each channel has its own configuration, so the metadata of each channel is reported separately.
- Each recording is processed by the GNSS-SDR software receiver right after it is made, and the outputs of every receiver stage are added under the recording in Section II (see Section 3).
- After 15 days of records from the last report. Rubem will send an automatic message to header of the Laboratory, Prof. Dr. Felix Antreich, via WhatsApp with the summarized information from all recorded metadata;

2 Recording calendar

Table 1 shows the days when recordings were made. The color of each day indicates how many recordings were made on that day: green means 2 recordings, blue means 1 recording, and no color means no recording was made.

October

M	T	W	T	F	S	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

Table 1: Recording calendar.

3 Validating the recordings with GNSS-SDR

A recording is only useful if a GNSS receiver can use it: find the satellites in it, track their signals, decode their navigation messages, and compute a position. To show that, every recording is processed with GNSS-SDR[2], an open-source software-defined GNSS receiver, and the output of each of its stages is added to the report of that recording (Section II). Figure 2 shows the data chain of GNSS-SDR. Each channel (one per satellite signal) runs its own acquisition, tracking and telemetry decoder; observables and PVT combine all of them.

[2] https://gnss-sdr.org, v0.0.20.

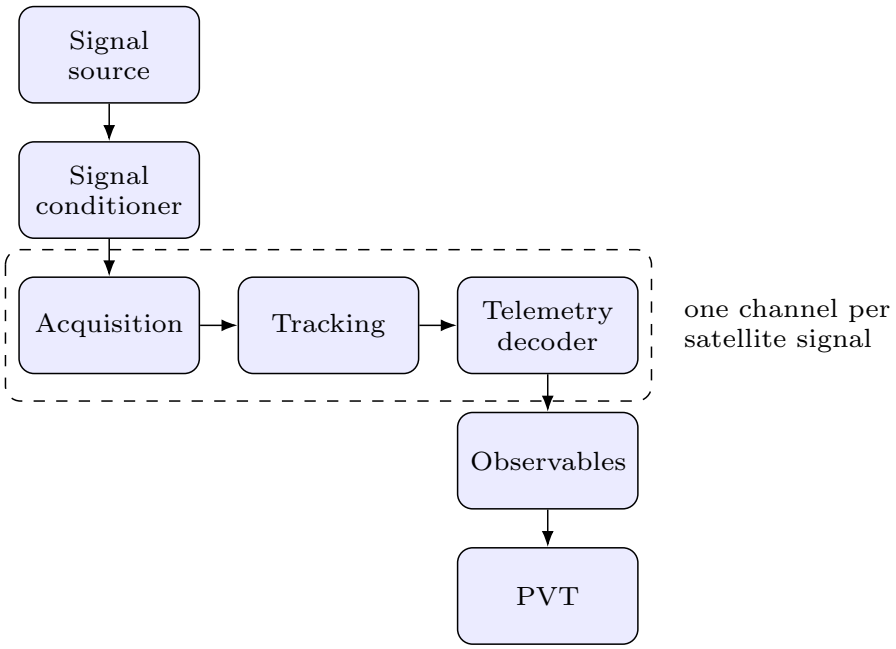


Figure 2: GNSS-SDR receiver chain.

Processing setup. Both channels of a recording are read by one `Multichannel_File_Signal_Source`, so the two bands advance in lock-step (with one independent source per band, one band runs ahead of the other and its measurements are discarded). That source reads both files at one rate: the 4 MS/s L1 C/A file of the L1/L5 recordings is upsampled 5 times (each sample repeated) to the 20 MS/s of L5 before GNSS-SDR. GNSS-SDR v0.0.20 computes its own position from L1 C/A only (broadcast ionosphere model): its dual-frequency solver ignores L2C altogether and was unstable with L5 on these recordings. The ionosphere-free, dual-frequency positioning is therefore computed outside GNSS-SDR's output, but using its observables and decoded ephemeris to generate our own single-point positioning (`gnss_spp.py`).

3.1 Signal source

Reads the recorded I/Q samples (here, the SigMF data files) and delivers them as a stream, one per channel. GNSS-SDR has no output of its own here, so the recording itself is examined.

- **Power spectral density** per channel: the shape of the receiver passband over a flat thermal-noise floor. GNSS signals are 15–20 dB below the noise, so they are invisible here; a narrow peak would be interference.
- **I/Q histograms**: should be Gaussian, centred on zero, and far from the integer limits (no clipping).

- **Table**: format, rate and gain per channel, I/Q standard deviation and mean, percentage of clipped values, and samples lost during the recording (overflows, zero-filled).

3.2 Signal conditioner

Adapts the samples to what the channels process: converts the file format (signed 8- or 16-bit interleaved I/Q) to complex floating point (data type adapter), optionally filters (input filter) and resamples (resampler).

- **Table** per channel: file format, data type adapter, filter, resampling, and rate before and after.

3.3 Acquisition

Detects which satellites are present and gives a first, coarse estimate of each one's code delay and Doppler shift, by correlating the signal with the satellite's spreading code over a two-dimensional grid of delays and Doppler frequencies, and comparing the peak with a detection threshold (set by the probability of false alarm). With dual-frequency assistance, the second band (L5 or L2C) is only searched for satellites already tracked on L1, around their L1 Doppler scaled to the second frequency.

- **Delay–Doppler map** (surface plot) of the strongest detection per band: a single sharp peak well above the floor.
- **Table**: search attempts, detections and detected satellites, and the best test statistic relative to the threshold. Attempts on satellites that are not in view, or that do not broadcast that band (L5 is only on Block IIF and III satellites, L2C on IIR-M, IIF and III), are rejected.

3.4 Tracking

Follows each detected signal with a delay-lock loop (code) and a phase-lock loop (carrier), producing every 1–20 ms the prompt correlator output (which carries the navigation data bits), the carrier-to-noise density ratio C/N_0, the Doppler, the code, and carrier phases.

- C/N_0 **and Doppler versus time** per satellite and band: C/N_0 of roughly 35–50 dB-Hz for an open-sky antenna, and smooth Doppler curves following the satellite motion.
- **Prompt I/Q scatter** of the strongest satellite per band: with the carrier phase locked, the energy sits on the in-phase axis in two clusters (the ±1 data bits); without phase lock it would be a ring.
- **Table** per satellite: median C/N_0, percentage of time phase-locked, losses of lock and time tracked.

3.5 Telemetry decoder

Synchronizes to the navigation-message bits and frames, checks their parity/CRC (Cyclic Redundancy Check), decodes the ephemeris (orbit and

satellite clock), and time-stamps every measurement with the satellite's transmit time (time of week). L1 C/A carries the legacy LNAV message (6-s subframes); L2C and L5 carry CNAV (12-s and 6-s messages).

- **Timeline** per band and satellite of when a valid transmit time was available.
- **Table**: messages decoded, ephemeris obtained, and when the first valid time of week came.

3.6 Observables

Aligns all channels on common receiver epochs and forms, for each satellite and band, the pseudorange (from transmit and receive times), the accumulated carrier phase and the Doppler. These are the measurements a positioning algorithm uses; GNSS-SDR also writes them to a RINEX observation file.

- **Availability** of valid pseudoranges per band and satellite over the recording.
- **Second band minus L1 pseudorange** per satellite: the same constant for every satellite (the receiver's inter-band hardware bias) plus the slowly varying, satellite-dependent ionospheric delay difference (metres). It proves that both bands measure the same satellites consistently.
- **Table** per satellite: common epochs, mean, standard deviation and drift of that difference.

3.7 PVT

Computes position, velocity and receiver clock (PVT) from the pseudoranges and the decoded ephemeris. The antenna position was surveyed: −23.209433°, −45.874875°, 596.42 m (WGS-84), so the errors below are true errors, not just scatter.

- **Horizontal error scatter and east/north/up error versus time** for GNSS-SDR's L1 C/A solution and for the dual-frequency ionosphere-free solution. Single-frequency single-point positioning is expected to stay within a few metres horizontally and ~10 m vertically; the iono-free combination removes the ionospheric bias but amplifies the code noise about three times.
- **Sky plot** of the satellites used.
- **Table**: mean error (accuracy), standard deviation (precision), 2-D and 3-D RMS, and 95% horizontal error.

Part II
Recording reports

4 Records on 08/10/2026

4.1 First record

Parameter	Channel 0	Channel 1
Start time	2026-10-08 22:26:37 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 2: Metadata for recording `x310_20261008T222630Z`, captured starting 2026-10-08T22:26:37.215640Z.

5 Records on 13/10/2026

5.1 First record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-13 21:57:56 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 3: Metadata for recording `x310_20261013T215749Z`, captured starting 2026-10-13T21:57:56.746726Z.

5.2 Second record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-13 22:08:41 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	
Sample rate	2.000 MS/s	
Sample type	signed complex 16-bit samples (I+Q), little-endian	
Gain	20.0 dB	20.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	2400.0 MB
Samples written	600,000,000	600,000,000
Lost samples	0	0
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	

Table 4: Metadata for recording `x310_20261013T220835Z`, captured starting 2026-10-13T22:08:41.589855Z.

6 Records on 22/10/2026

6.1 First record

Parameter	**Channel 0**	**Channel 1**
Start time	2026-10-22 19:00:06 UTC	
Frequency	1575.420 MHz (L1)	1176.450 MHz (L5)
Duration	300.0 s	300.0 s
Sample rate	4.000 MS/s	20.000 MS/s
Sample type	signed complex 8-bit samples (I+Q), converted on the host from the X310's sc16	
Gain	25.0 dB	25.0 dB
Antenna	RX1	RX2
File size	2400.0 MB	12000.0 MB
Samples written	1,200,000,000	6,000,000,000
Lost samples	2912	14924
Clock source	external	
Time source	internal	
Motherboard	USRP X310	
Daughterboard	TwinRX RX0	TwinRX RX1
Recorder	UHD Python API	
GNSS-SDR configuration	`l1ca8bits4msps_l5_8bits20msps.conf`	

Table 5: Metadata for recording `x310_20261022T190000Z`, captured starting 2026-10-22T19:00:06.973177Z.

6.1.1 Signal source

Raw I/Q samples of each channel analysed within the window [10, 10.5] s after the start recording instant.

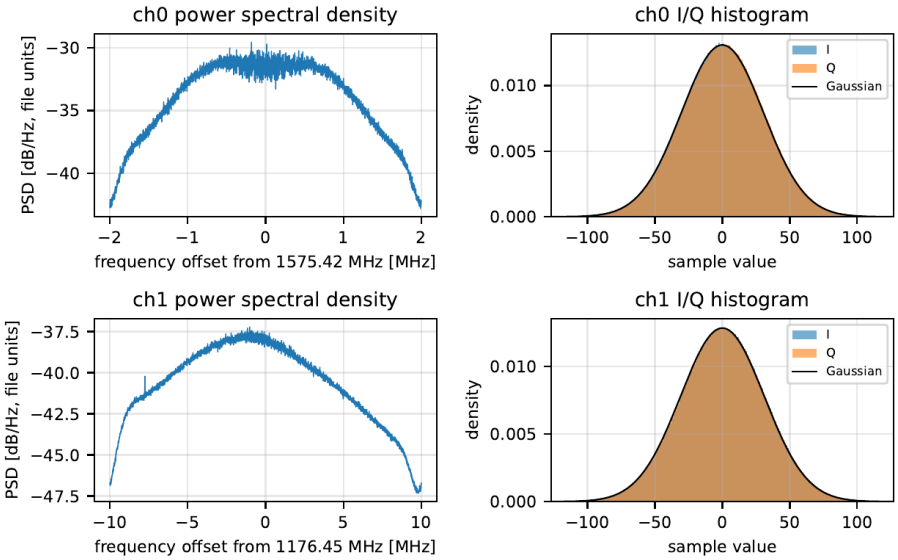


Figure 3: Signal source: power spectral density (the receiver passband over a flat thermal-noise floor; GNSS signals are below the noise, so only interference would stand out) and I/Q histograms (Gaussian noise, centred, not clipped).

Channel	**ch0**	**ch1**
Frequency [MHz]	1575.42	1176.45
Rate [MS/s]	4	20
Format	ci8	ci8
Gain [dB]	25	25
Std I / Q	30.4 / 30.5	31.1 / 31.1
Mean I / Q	-0.26 / 0.20	0.26 / -0.07
Clipped [%]	0.003	0.004
Lost samples	2,912	14,924

Table 6: Signal source: sample statistics per channel.

6.1.2 Signal conditioner

How each recorded channel is turned into the complex baseband stream the channels process.

Channel	ch0 (L1 C/A)	ch1 (L5)
File format	ci8	ci8
Source item type	ibyte	ibyte
Data type adapter	Ibyte_To_Complex	Ibyte_To_Complex
Input filter	Pass_Through	Pass_Through
Resampling	×5 zero-order hold (offline)	Pass_Through
Rate [MS/s]	4 → 20	20 → 20

Table 7: Signal conditioner: recorded format and rate to gnss-sdr's internal `gr_complex` stream; both bands are read in lockstep by one `Multichannel_File_Signal_Source`.

6.1.3 Acquisition

Two-dimensional search over code delay and Doppler (one channel per band dumped; the others behave alike).

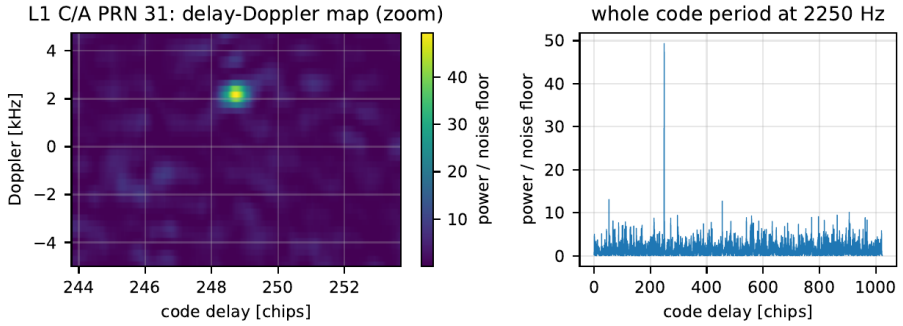


Figure 4: Acquisition, L1 C/A: delay–Doppler search of PRN 31; one sharp peak (test statistic 71 vs threshold 47) at 2250 Hz and 248.7 chips.

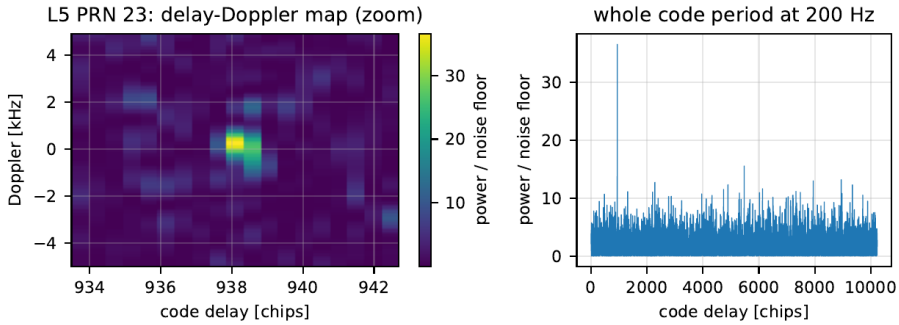


Figure 5: Acquisition, L5: delay–Doppler search of PRN 23; one sharp peak (test statistic 51 vs threshold 49) at -1587 Hz and 938.1 chips.

Band	Attempts	Detections	Detected PRNs	Best statistic / threshold
L1 C/A	11	1	31	1.5
L5	115	1	23	1.0

Table 8: Acquisition: search attempts of the dumped channel. Attempts on satellites that are not in view (or, for L5/L2C, do not broadcast that signal) are rejected below the threshold.

6.1.4 Tracking

Code (DLL) and carrier (PLL) loops per satellite.

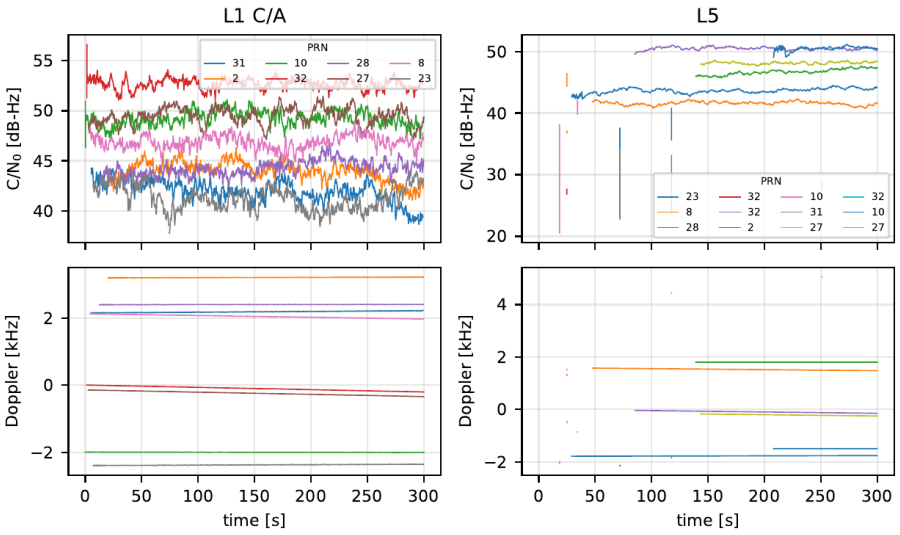


Figure 6: Tracking: carrier-to-noise density ratio and carrier Doppler per satellite, while phase-locked (smooth Doppler curves = loops following the real satellite motion).

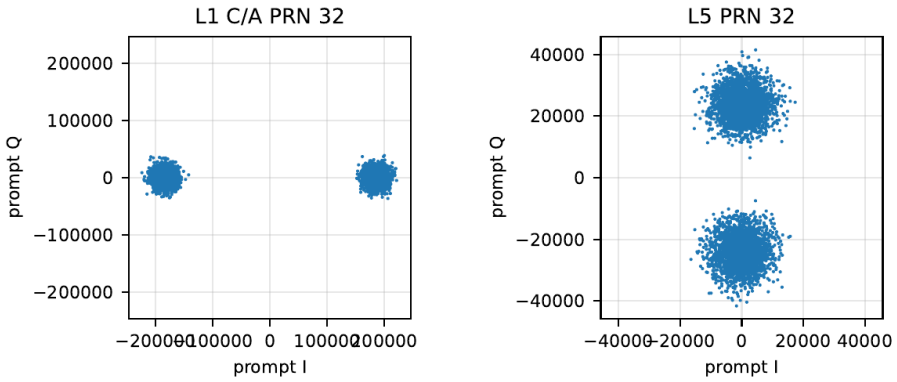


Figure 7: Tracking: prompt correlator outputs of the strongest satellite per band. Two tight clusters on one axis (the ±1 data bits; GNSS-SDR's L5 tracking puts the data component on the Q axis) mean the PLL is phase-locked; a ring would mean it is not.

Band	Ch.	PRN	Median C/N_0 [dB-Hz]	Phase-locked [%]	Losses of lock	Tracked [s]
L1 C/A	0	31	42.7	93	0	561
L1 C/A	1	2	43.5	95	0	28
L1 C/A	2	10	49.2	100	0	512
L1 C/A	3	32	52.9	100	0	30
L1 C/A	4	28	43.8	93	0	526
L1 C/A	5	27	49.0	94	0	528
L1 C/A	6	8	47.0	92	0	30
L1 C/A	7	23	41.9	92	0	530
L5	8	23	43.5	95	0	372
L5	9	8	41.7	67	0	374
L5	10	28	46.4	58	1	279
L5	10	32	28.9	0	1	16
L5	11	32	50.4	66	0	327
L5	12	2	28.8	0	1	16
L5	13	10	29.2	1	5	32
L5	13	31	28.7	0	5	48
L5	14	27	48.2	56	1	279
L5	14	32	28.9	0	1	16
L5	15	10	32.2	36	3	51
L5	15	27	28.8	0	3	16

Table 9: Tracking: per channel and satellite (phase-locked: carrier lock test > 0.8).

6.1.5 Telemetry decoder

Navigation-message decoding: bit/frame synchronization, parity/CRC, the ephemeris, and the satellite transmit time (time of week) of every measurement.

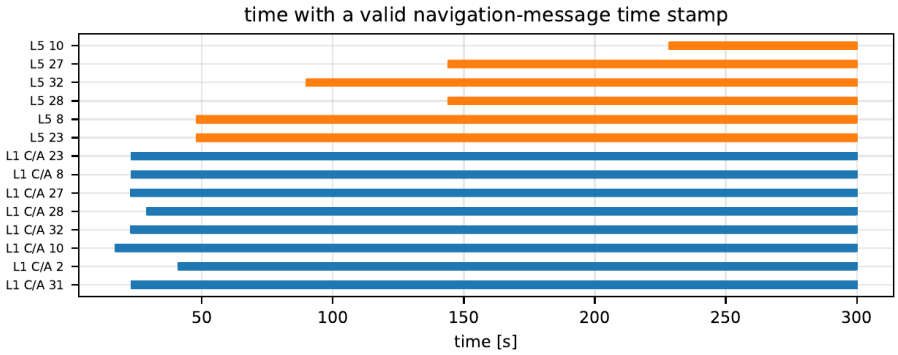


Figure 8: Telemetry decoder: per band and satellite, when the decoder delivered a valid transmit time (the first one comes after the first complete frame/message).

Band	PRN	Messages decoded	Ephemeris	First valid at [s]	TOW [s]
L1 C/A	2	44	yes	41.1	414066.0
L1 C/A	8	46	yes	23.1	414048.0
L1 C/A	10	48	yes	17.1	414042.0
L1 C/A	23	46	yes	23.1	414048.0
L1 C/A	27	47	yes	23.1	414048.0
L1 C/A	28	44	yes	29.1	414054.0
L1 C/A	31	47	yes	23.1	414048.0
L1 C/A	32	47	yes	23.1	414048.0
L5	8	25	yes	48.1	414073.0
L5	10	7	yes	228.4	414253.3
L5	23	31	yes	48.1	414073.0
L5	27	15	yes	144.1	414169.0
L5	28	19	yes	144.1	414169.0
L5	32	22	yes	90.0	414114.9

Table 10: Telemetry decoder: L1 C/A LNAV subframes and L2C/L5 CNAV messages decoded, ephemeris obtained, and the first valid time of week.

6.1.6 Observables

Pseudorange, carrier phase and Doppler of every channel at common receiver epochs.

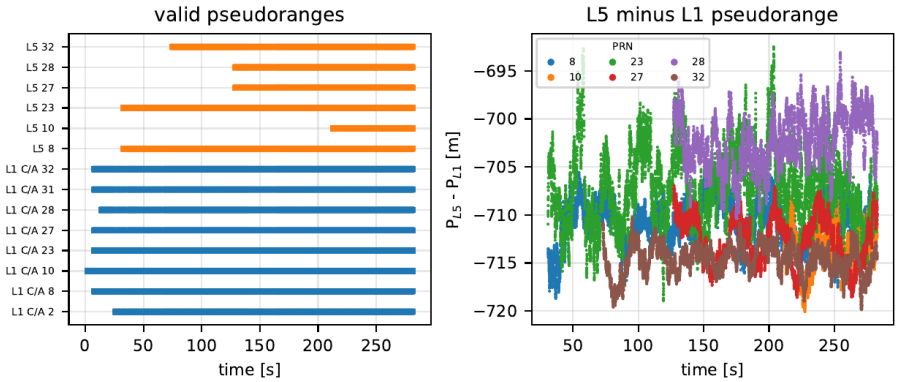


Figure 9: Observables: epochs with a valid pseudorange per band and satellite, and the difference between the two bands' pseudoranges of the same satellite (a constant receiver inter-band bias, the same for every satellite, plus the slowly varying ionospheric delay).

PRN	Common epochs	of run [%]	Mean [m]	Std [m]	Drift [m/s]
8	12228	82	-710.75	2.19	0.0014
10	3563	24	-713.39	2.20	0.0208
23	12231	82	-707.12	3.91	0.0006
27	7627	51	-712.62	2.45	-0.0107
28	7628	51	-702.35	2.80	0.0221
32	10330	69	-714.67	1.54	-0.0003

Table 11: Observables: $P_{L5} - P_{L1}$ per satellite at common epochs (similar means = shared receiver bias; std = code noise of both bands).

6.1.7 PVT

Position from the pseudoranges and the decoded ephemeris, compared with the surveyed antenna position (-23.209433°, -45.874875°, 596.42 m).

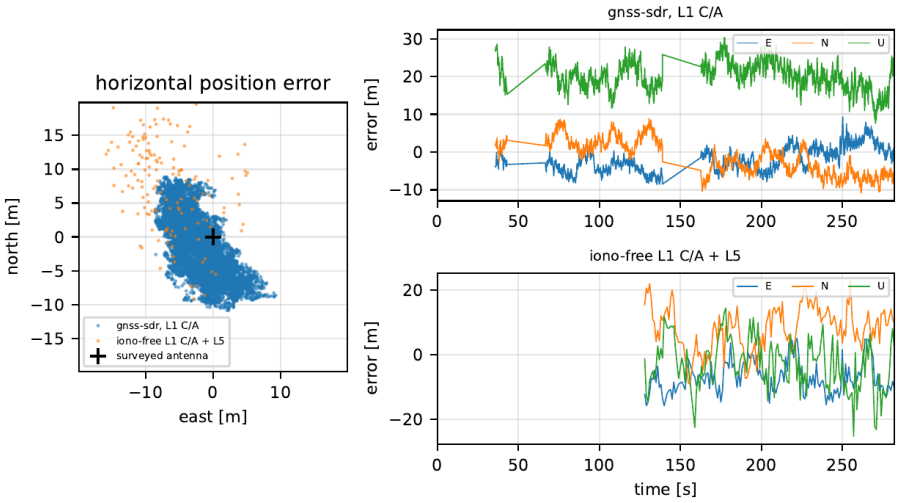


Figure 10: PVT: position error with respect to the surveyed antenna position. L1 C/A: gnss-sdr's own solution (broadcast ionosphere model). Iono-free: computed from gnss-sdr's observables and decoded ephemeris (gnss_spp.py), removing the ionospheric delay with the second band (pseudoranges carrier-smoothed over 100 s).

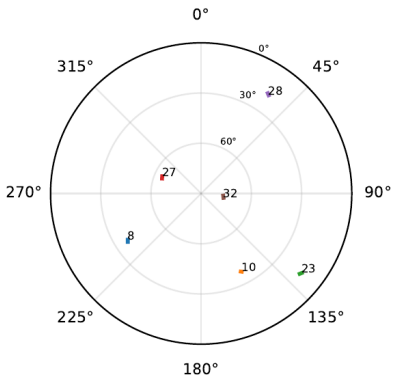


Figure 11: PVT: sky plot of the satellites in the dual-frequency solution (azimuth, elevation) during the recording.

Solution	**gnss-sdr, L1 C/A**	**iono-free L1 C/A + L5**
Solutions	10332	151
Run with fix [%]	72	54
Mean E / N / U [m]	-2.0 / -1.5 / 20.0	-7.0 / 8.2 / -2.9
Std E / N / U [m]	3.2 / 4.5 / 3.6	5.0 / 6.7 / 7.9
RMS 2-D [m]	6.1	13.6
RMS 3-D [m]	21.2	16.0
95% horiz. [m]	9.3	20.6

Table 12: PVT: position error statistics with respect to the surveyed antenna (mean = accuracy, std = precision; % of the run's seconds with a position; gnss-sdr outputs several solutions per second, the iono-free one is computed once per second).